

Retail Sales & Inventory Intelligence System Using Excel, SQL, and Power BI

Problem Statement:

Design and implement a comprehensive data analysis solution for a retail company that sells mobile phones and laptops. The company operates multiple stores and wants to analyze its end-to-end operations from order processing to inventory management and sales performance—across regions, stores, products, and staff. The company’s management seeks to extract insights that can improve sales strategies, identify top-selling products, understand customer preferences, optimize inventory, and monitor staff performance.

Given Table:

The dataset consists of 9 tables including brands, customers, staffs, orders, order_items, stores, stocks, products, categories.

Phase-1 -Cleaned Excel :

1. Brand.csv

Brand_Id	Brand_Name
1	Electra
2	Haro
3	Heller
4	Pure Cycles
5	Ritchey
6	Strider
7	Sun Bicycles
8	Surly
9	Trek

2.Categories.csv

Category_Id	Category_Name
1	Children Bicycles
2	Comfort Bicycles
3	Cruisers Bicycles
4	Cyclocross Bicycles
5	Electric Bikes
6	Mountain Bikes
7	Road Bikes

3. Customers.csv

Customer_Id	First_Name	Last_Name	Phone	Email	Street	City	State	Zip_Code
1	Debra	Burks	Not Available	debra.burks@yahoo.com	9273 Thorne Ave.	Orchard Park	NY	14127
2	Kasha	Todd	Not Available	kasha.todd@yahoo.com	910 Vine Street	Campbell	CA	95008
3	Tameka	Fisher	Not Available	tameka.fisher@aol.com	769C Honey Creek St.	Redondo Beach	CA	90278
4	Daryl	Spence	Not Available	daryl.spence@aol.com	988 Pearl Lane	Uniondale	NY	11553
5	Charolette	Rice	(916) 381-6003	charolette.rice@msn.com	107 River Dr.	Sacramento	CA	95820
6	Lyndsey	Bean	Not Available	lyndsey.bean@hotmail.com	769 West Road	Fairport	NY	14450
7	Latasha	Hays	(716) 986-3359	latasha.hays@hotmail.com	7014 Manor Station Rd.	Buffalo	NY	14215
8	Jacqueline	Duncan	Not Available	jacqueline.duncan@yahoo.com	15 Brown St.	Jackson Heights	NY	11372
9	Genoveva	Baldwin	Not Available	genoveva.baldwin@msn.com	8550 Spruce Drive	Port Washington	NY	11050
10	Pamela	Newman	Not Available	pamela.newman@gmail.com	476 Chestnut Ave.	Monroe	NY	10950
11	Deshawn	Mendoza	Not Available	deshawn.mendoza@yahoo.com	8790 Cobblestone Street	Monsey	NY	10952
12	Robby	Sykes	(516) 583-7761	robby.sykes@hotmail.com	486 Rock Maple Street	Hempstead	NY	11550
13	Lashawn	Ortiz	Not Available	lashawn.ortiz@msn.com	27 Washington Rd.	Longview	TX	75604
14	Garry	Espinoza	Not Available	garry.espinoza@hotmail.com	7858 Rockaway Court	Forney	TX	75126
15	Linnie	Branch	Not Available	linnie.branch@gmail.com	314 South Columbia Ave.	Plattsburgh	NY	12901
16	Emmitt	Sanchez	(212) 945-8823	emmitt.sanchez@hotmail.com	461 Squaw Creek Road	New York	NY	10002
17	Caren	Stephens	Not Available	caren.stephens@msn.com	914 Brook St.	Scarsdale	NY	10583
18	Georgetta	Hardin	Not Available	georgetta.hardin@aol.com	474 Chapel Dr.	Canandaigua	NY	14424
19	Lizette	Stein	Not Available	lizette.stein@yahoo.com	19 Green Hill Lane	Orchard Park	NY	14127
20	Aleta	Shepard	Not Available	aleta.shepard@aol.com	684 Howard St.	Sugar Land	TX	77478
21	Tobie	Little	Not Available	tobie.little@gmail.com	10 Silver Spear Dr.	Victoria	TX	77904
22	Adelle	Larsen	Not Available	adelle.larsen@gmail.com	683 West Kirkland Dr.	East Northport	NY	11731

4. Order_items.csv:

Order_Id	Item_Id	Product_Id	Quantity	List_Price	Discount	Net_Revenue
1	1	20	1	₹ 600	20%	₹ 480
1	2	8	2	₹ 1,800	7%	₹ 3,348
1	3	10	2	₹ 1,549	5%	₹ 2,943
1	4	16	2	₹ 600	5%	₹ 1,140
1	5	4	1	₹ 2,900	20%	₹ 2,320
2	1	20	1	₹ 600	7%	₹ 558
2	2	16	2	₹ 600	5%	₹ 1,140
3	1	3	1	₹ 1,000	5%	₹ 950
3	2	20	1	₹ 600	5%	₹ 570
4	1	2	2	₹ 750	10%	₹ 1,350
5	1	10	2	₹ 1,549	5%	₹ 2,943
5	2	17	1	₹ 429	7%	₹ 399
5	3	26	1	₹ 600	7%	₹ 558
6	1	18	1	₹ 449	7%	₹ 418
6	2	12	2	₹ 550	5%	₹ 1,045
6	3	20	1	₹ 600	10%	₹ 540
6	4	3	2	₹ 1,000	7%	₹ 1,860
6	5	9	2	₹ 3,000	7%	₹ 5,580
7	1	15	1	₹ 530	7%	₹ 493
7	2	3	1	₹ 1,000	10%	₹ 900
7	3	17	2	₹ 429	10%	₹ 772
8	1	22	1	₹ 270	5%	₹ 256

5.Orders.csv:

Order_Id	Customer_Id	Order_Status	Order_Date	Required_Date	Shipped_Date	Store_Id	Staff_Id	Order_Status_Label
1	259	4	1/1/2016	1/3/2016	1/3/2016	1	2	Completed
2	1212	4	1/1/2016	1/4/2016	1/3/2016	2	6	Completed
3	523	4	1/2/2016	1/5/2016	1/3/2016	2	7	Completed
4	175	4	1/3/2016	1/4/2016	1/5/2016	1	3	Completed
5	1324	4	1/3/2016	1/6/2016	1/6/2016	2	6	Completed
6	94	4	1/4/2016	1/7/2016	1/5/2016	2	6	Completed
7	324	4	1/4/2016	1/7/2016	1/5/2016	2	6	Completed
8	1204	4	1/4/2016	1/5/2016	1/5/2016	2	7	Completed
9	60	4	1/5/2016	1/8/2016	1/8/2016	1	2	Completed
10	442	4	1/5/2016	1/6/2016	1/6/2016	2	6	Completed
11	1326	4	1/5/2016	1/8/2016	1/7/2016	2	7	Completed
12	91	4	1/6/2016	1/8/2016	1/9/2016	1	2	Completed
13	873	4	1/8/2016	1/11/2016	1/11/2016	2	6	Completed
14	258	4	1/9/2016	1/11/2016	1/12/2016	1	3	Completed
15	450	4	1/9/2016	1/10/2016	1/12/2016	2	7	Completed
16	552	4	1/12/2016	1/15/2016	1/15/2016	1	3	Completed
17	1175	4	1/12/2016	1/14/2016	1/14/2016	1	3	Completed
18	541	4	1/14/2016	1/17/2016	1/15/2016	1	3	Completed
19	696	4	1/14/2016	1/17/2016	1/16/2016	1	2	Completed
20	923	4	1/14/2016	1/16/2016	1/17/2016	1	2	Completed
21	1250	4	1/15/2016	1/16/2016	1/18/2016	2	6	Completed
22	1035	4	1/16/2016	1/18/2016	1/17/2016	1	2	Completed

6. Products.csv:

Product_Id	Product_Name	Brand_Id	Category_Id	Model_Year	List_Price
1	Trek 820 - 2016	9	6	2016	₹ 380
2	Ritchey Timberwolf Frameset - 2016	5	6	2016	₹ 750
3	Surly Wednesday Frameset - 2016	8	6	2016	₹ 1,000
4	Trek Fuel EX 8 29 - 2016	9	6	2016	₹ 2,900
5	Heller Shagamaw Frame - 2016	3	6	2016	₹ 1,321
6	Surly Ice Cream Truck Frameset - 2016	8	6	2016	₹ 470
7	Trek Slash 8 27.5 - 2016	9	6	2016	₹ 4,000
8	Trek Remedy 29 Carbon Frameset - 2016	9	6	2016	₹ 1,800
9	Trek Conduit+ - 2016	9	5	2016	₹ 3,000
10	Surly Straggler - 2016	8	4	2016	₹ 1,549
11	Surly Straggler 650b - 2016	8	4	2016	₹ 1,681
12	Electra Townie Original 21D - 2016	1	3	2016	₹ 550
13	Electra Cruiser 1 (24-Inch) - 2016	1	3	2016	₹ 270
14	Electra Girl's Hawaii 1 (16-inch) - 2015/2016	1	3	2016	₹ 270
15	Electra Moto 1 - 2016	1	3	2016	₹ 530
16	Electra Townie Original 7D EQ - 2016	1	3	2016	₹ 600
17	Pure Cycles Vine 8-Speed - 2016	4	3	2016	₹ 429
18	Pure Cycles Western 3-Speed - Women's - 2015/2016	4	3	2016	₹ 449
19	Pure Cycles William 3-Speed - 2016	4	3	2016	₹ 449
20	Electra Townie Original 7D EQ - Women's - 2016	1	3	2016	₹ 600
21	Electra Cruiser 1 (24-Inch) - 2016	1	1	2016	₹ 270
22	Electra Girl's Hawaii 1 (16-inch) - 2015/2016	1	1	2016	₹ 270

7. Staff.csv:

Staff_Id	First_Name	Last_Name	Email	Phone	Active	Store_Id	Manager_Id
1	Fabiola	Jackson	fabiola.jackson@bikes.shop	(831) 555-5554	1	1	NULL
2	Mireya	Copeland	mireya.copeland@bikes.shop	(831) 555-5555	1	1	1
3	Genna	Serrano	genna.serrano@bikes.shop	(831) 555-5556	1	1	2
4	Virgie	Wiggins	virgie.wiggins@bikes.shop	(831) 555-5557	1	1	2
5	Jannette	David	jannette.david@bikes.shop	(516) 379-4444	1	2	1
6	Marcelene	Boyer	marcelene.boyer@bikes.shop	(516) 379-4445	1	2	5
7	Venita	Daniel	venita.daniel@bikes.shop	(516) 379-4446	1	2	5
8	Kali	Vargas	kali.vargas@bikes.shop	(972) 530-5555	1	3	1
9	Layla	Terrell	layla.terrell@bikes.shop	(972) 530-5556	1	3	7
10	Bernardine	Houston	bernardine.houston@bikes.shop	(972) 530-5557	1	3	7

8. Stores.csv:

Store_Id	Store_Name	Phone	Email	Street	City	State	Zip_Code
1	Santa Cruz Bikes	(831) 476-4321	santacruz@bikes.shop	3700 Portola Drive	Santa Cruz	CA	95060
2	Baldwin Bikes	(516) 379-8888	baldwin@bikes.shop	4200 Chestnut Lane	Baldwin	NY	11432
3	Rowlett Bikes	(972) 530-5555	rowlett@bikes.shop	8000 Fairway Avenue	Rowlett	TX	75088

9. Stocks.csv:

Store_Id	Product_Id	Quantity
1	1	27
1	2	5
1	3	6
1	4	23
1	5	22
1	6	0
1	7	8
1	8	0
1	9	11
1	10	15
1	11	8
1	12	16
1	13	13
1	14	8
1	15	3
1	16	4
1	17	2
1	18	16
1	19	4
1	20	26
1	21	24
1	22	29

Key Business Findings from Raw Data in Excel:

- **Date Range:**
Data covers transactions from **2016-01-01 to 2018-12-28** (3 years of business data).
- **Order Status Breakdown:**
 - Status 4 (Completed) - **1,445 orders**
 - Status 1 (Pending) - **62 orders**
 - Status 2 (Processing) - **63 orders**
 - Status 3 (Rejected) - **45 orders**
- **Delayed Shipments:**
 - **458 orders** were shipped after the required delivery date.
 - Indicates a major operational/logistics issue affecting delivery performance.
- **NULL shipped_date Values:**
 - **170 rows** contain NULL values in shipped_date.
 - All belong to Pending, Processing, or Rejected orders.
 - This is expected business behavior, not a data quality issue.
- **Discount Pattern:**
 - Every item has a discount between **5% and 20%**.
 - No products were sold without discounts.
- **Customer Phone Data Quality:**
 - **87.6% of customer phone numbers are missing.**
 - Phone data is not reliable for analysis or customer outreach.

Phase 2- SQL:

Connected the CSV files to MySQL using the MySQL Connector and imported the data into SQL tables for analysis. As part of the data analyst workflow, performed data validation, checked for missing values, analyzed order statuses, shipment delays, discount patterns, and verified data consistency before proceeding with further analysis and dashboard creation.

Q1: Revenue by Store.

store_name	total_revenue
Baldwin Bikes	5215746.64
Santa Cruz Bikes	1605821.56
Rowlett Bikes	867541.51

Q2: Revenue by brand.

brand_name	total_revenue
Trek	4602752.41
Electra	1205318.10
Surly	949506.31
Sun Bicycles	341994.16
Haro	185384.19
Heller	171458.93
Pure Cycles	149476.34
Ritchey	78898.82
Strider	4320.45

Q3: Revenue by category

category_name	total_revenue
Mountain Bikes	2715077.72
Road Bikes	1665097.93
Cruisers Bicycles	995030.86
Electric Bikes	916684.45
Cyclocross Bicycles	711011.57
Comfort Bicycles	394019.27
Children Bicycles	292187.91

Q4: Top 10 best-selling products

product_name	total_quantity_sold	total_revenue
Electra Cruiser 1 (24-Inch) - 2016	296	72070.84
Electra Townie Original 7D EQ - 2016	290	155169.10
Electra Townie Original 21D - 2016	289	143393.11
Electra Girl's Hawaii 1 (16-inch) - 2015/2016	269	64513.81
Surly Ice Cream Truck Frameset - 2016	167	70371.43
Trek Slash 8 27.5 - 2016	154	555558.46
Electra Girl's Hawaii 1 (20-inch) - 2015/2016	154	41011.46
Surly Straggler 650b - 2016	151	226765.39
Electra Townie Original 7D - 2015/2016	148	66568.52
Surly Straggler - 2016	147	203507.62

Q5: Staff performance - orders handled + revenue

staff_name	total_orders	total_revenue
Marcelene Boyer	553	2624118.25
Venita Daniel	540	2591628.39
Genna Serrano	184	853286.55
Mireya Copeland	164	752535.01
Kali Vargas	88	463917.91
Layla Terrell	86	403623.60

Q6: Monthly sales trend (2016–2018)

order_year	order_month	monthly_revenue
2016	1	215146.23
2016	2	156112.11
2016	3	180600.13
2016	4	167143.90
2016	5	205269.82
2016	6	210561.95
2016	7	199556.61
2016	8	225657.16
2016	9	273091.37
2016	10	212077.86
2016	11	182329.24
2016	12	199829.79
2017	1	285616.26
2017	2	312923.47
2017	3	308911.60
2017	4	227290.63
2017	5	268232.99
2017	6	378865.34
2017	7	229995.14
2017	8	290553.17
2017	9	293405.03
2017	10	310328.02
2017	11	281577.65
2017	12	259505.76
2018	1	381429.86
2018	2	200657.90
2018	3	363990.72
2018	4	817921.28
2018	6	188.99
2018	7	11337.89
2018	8	8377.80
2018	9	8963.96
2018	10	3781.12
2018	11	11362.00
2018	12	6516.96

Q7: Delayed shipment analysis by store with delay rate %

store_name	total_orders	delayed_orders	delay_rate_percent
Santa Cruz Bikes	348	104	29.89
Baldwin Bikes	1093	317	29.00
Rowlett Bikes	174	37	21.26

Q8: Top 15 customers by order frequency + lifetime value

customer_id	customer_name	total_orders	lifetime_value
94	Sharyn Hopkins	2	34807.93
10	Pamelia Newman	3	33634.24
75	Abby Gamble	2	32802.99
6	Lyndsey Bean	3	32675.06
16	Emmitt Sanchez	3	31925.87
73	Melanie Hayes	2	31913.68
1	Debra Burks	3	27888.17
61	Elinore Aguilar	3	25636.43
93	Corrina Sawyer	2	25612.69
122	Shena Carter	1	24890.62
1224	Abram Copeland	1	24607.02
9	Genoveva Baldwin	3	24433.87
12	Robby Sykes	3	24305.13
3	Tameka Fisher	3	24051.51
85	Teofila Fischer	2	23195.06

Q9: Customer distribution by state

state	total_customers
NY	1019
CA	284
TX	142

Q10: Inventory low stock alert (Out of Stock / Low / Adequate)

store_name	product_name	quantity	stock_status
Santa Cruz Bikes	Surly Ice Cream Truck Frameset - 2016	0	Out of Stock
Santa Cruz Bikes	Trek Remedy 29 Carbon Frameset - 2016	0	Out of Stock
Santa Cruz Bikes	Trek Farley Alloy Frameset - 2017	0	Out of Stock
Santa Cruz Bikes	Trek Fuel EX 5 27.5 Plus - 2017	0	Out of Stock
Santa Cruz Bikes	Haro Shredder 20 - 2017	0	Out of Stock
Santa Cruz Bikes	Trek Emonda SLR 6 - 2018	0	Out of Stock
Santa Cruz Bikes	Surly Pack Rat - 2018	0	Out of Stock
Santa Cruz Bikes	Surly Straggler - 2018	0	Out of Stock
Santa Cruz Bikes	Electra Townie Original 3i EQ Ladies' - 2018	0	Out of Stock
Santa Cruz Bikes	Electra Townie Original 1 - 2018	0	Out of Stock
Baldwin Bikes	Electra Girl's Hawaii 1 (16-inch) - 2015/2016	0	Out of Stock
Baldwin Bikes	Trek Remedy 9.8 - 2017	0	Out of Stock
Baldwin Bikes	Trek Precaliber 24 (21-Speed) - Girls - 2017	0	Out of Stock
Baldwin Bikes	Trek CrossRip 1 - 2018	0	Out of Stock
Baldwin Bikes	Trek Domane SLR Frameset - 2018	0	Out of Stock
Baldwin Bikes	Trek Domane SL 6 Disc - 2018	0	Out of Stock
Baldwin Bikes	Electra Townie Go! 8i - 2017/2018	0	Out of Stock
Baldwin Bikes	Electra Townie Commute Go! - 2018	0	Out of Stock
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Q11: Year-over-year revenue by store (pivot-style)

store_name	order_year	total_revenue
Baldwin Bikes	2016	1590597.24
Baldwin Bikes	2017	2479686.19
Baldwin Bikes	2018	1145463.21
Rowlett Bikes	2016	271285.91
Rowlett Bikes	2017	404611.24
Rowlett Bikes	2018	191644.36
Santa Cruz Bikes	2016	565493.02
Santa Cruz Bikes	2017	562907.63
Santa Cruz Bikes	2018	477420.91

Q12: Discount tier impact on revenue

discount_tier	total_orders	total_revenue	avg_revenue
6-10%	2314	3871618.52	1673.13
0-5%	1205	2138328.59	1774.55
16-20%	1203	1679162.60	1395.81

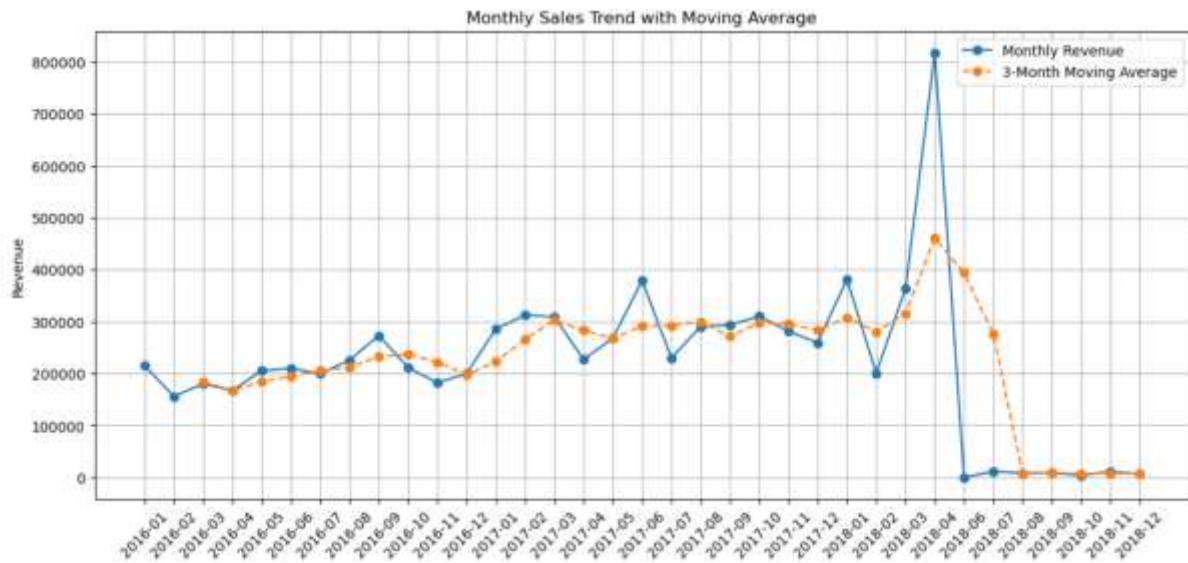
Phase 3 – Python:

Data preprocessing was completed successfully to prepare the dataset for analysis and dashboard creation. The process included handling missing values, validating data types, checking duplicate records, standardizing column formats, and cleaning inconsistent entries. NULL values in shipment-related fields were verified against order statuses to ensure data accuracy. Additional preprocessing steps included date formatting, discount validation, and preparing the dataset for SQL analysis, Python-based EDA, and visualization in Power BI.

Customer Lifetime Value:

	customer_id	first_name	last_name	total_orders	lifetime_value
0	1	Debra	Burks	3	27888.17
1	2	Kasha	Todd	3	19329.07
2	3	Tameka	Fisher	3	24051.51
3	4	Daryl	Spence	3	21150.88
4	5	Charolette	Rice	3	17520.28

Moving Average Trend:



The monthly sales trend analysis revealed a steady increase in revenue from 2016 to early 2018, indicating positive business growth. A major sales spike was observed in March 2018, likely driven by seasonal demand or promotional activity. However, revenue sharply declined afterward, suggesting potential data quality or transaction recording issues. A 3 month moving average was applied to smooth fluctuations and better understand long-term sales behavior.

Customer Segmentation – Clustering:

	customer_id	first_name	last_name	total_orders	lifetime_value	segment
0	1	Debra	Burks	3	27888.17	1
1	2	Kasha	Todd	3	19329.07	1
2	3	Tameka	Fisher	3	24051.51	1
3	4	Daryl	Spence	3	21150.88	1
4	5	Charolette	Rice	3	17520.28	1

Segment 0 - High value customers, Segment 1 - Medium customers, Segment 3 - Low value customers

Phase 4 -Power Bi Data Visualization:



Overall Business Insights – Retail Sales & Inventory Intelligence System

1. Strong Revenue Growth

- The business showed a consistent increase in sales from 2016 to early 2018.
- Monthly revenue trends indicate healthy customer demand and operational stability.
- Peak revenue periods suggest successful sales strategies and strong market performance.

2. High Sales Concentration in Certain Periods

- Sales spikes during specific months indicate:
 - seasonal demand,
 - promotional success,
 - or bulk purchasing behavior.
- These periods can be leveraged for future marketing campaigns and inventory planning.

3. Customer Purchasing Behavior

- A small group of customers contributed significantly to total revenue.
- Customer segmentation revealed:
 - low-value customers,
 - regular customers,
 - premium/high-value customers.

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- High-value customers can be targeted through loyalty programs and personalized offers.

4. Store Performance Variation

- Some stores consistently outperformed others in revenue generation.
- Underperforming stores may require:
 - better stock management,
 - regional promotions,
 - operational improvements.

5. Product & Category Performance

- Certain product categories generated the majority of sales revenue.
- Fast-moving products showed strong and consistent demand.
- Slow-moving inventory may increase storage and operational costs.

6. Sales Trend Stability

- The moving average analysis helped identify the long-term business trend.
- Despite monthly fluctuations, the overall sales direction remained positive until the sudden decline in 2018.

7. Potential Data Quality Issue

- A sharp drop in revenue after the major 2018 spike suggests:
 - missing transactions,
 - ETL/data pipeline issues,
 - incomplete database updates.
- Data validation and monitoring processes should be improved.

8. Operational & Strategic Insights

The analytics system enables:

- better inventory planning,
- demand forecasting,
- customer behavior analysis,
- regional sales monitoring,
- staff/store performance tracking.

9. Dashboard & Reporting Impact

The Power BI dashboard provides:

- real-time KPI tracking,
- interactive filtering,
- drill-down analysis,
- executive-level business visibility.

This improves decision-making speed and business intelligence capabilities.

Final Business Conclusion

The Retail Sales & Inventory Intelligence System successfully transformed raw transactional data into actionable business insights. The project identified sales trends, customer segments, high-performing stores, and operational opportunities while enabling interactive reporting through Power BI. The solution supports data-driven decision-making, inventory optimization, and strategic business planning.